Remote Sensing Applications: Society and Environment 41 (2026) 101951



Contents lists available at ScienceDirect

Remote Sensing Applications: Society and Environment

journal homepage: www.elsevier.com/locate/rsase





Natural language-driven water quality inversion using LLMs and cloud platforms

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ARTICLE INFO

Keywords:
Water quality remote sensing inversion
Large language model (LLM)
Google Earth Engine (GEE)
Retrieval-augmented generation (RAG)
Natural language processing (NLP)

ABSTRACT

This study presents a water quality retrieval framework that combines natural language-driven large language models (LLMs) with Google Earth Engine, aiming to simplify traditional remote sensing monitoring by reducing its dependence on specialized knowledge and technical procedures. Through the integration of a retrieval-augmented generation (RAG) mechanism, the framework achieves strong performance in complex water quality tasks, with a task parsing consistency of 0.92, a spatial matching rate of 94%, and a temporal matching rate of 93%, markedly outperforming zero-shot and few-shot approaches. Evaluation on water quality semantic tasks shows that RAG achieves an accuracy of 0.72, a recall of 0.91, and an F1-score of 0.79 in classification tasks, as well as a mean reciprocal rank of 0.85, top-1 accuracy of 0.73, and top-3 accuracy of 0.97 in ranking tasks, demonstrating the effectiveness of the framework in identifying key spatiotemporal constraints and prioritizing high-quality remote sensing data. The RAG-enhanced Qwen2.5-Coder model attains an executable code rate of 87%, substantially improving the translation of natural language queries into operational water quality retrieval workflows. When combined with visualization and LLM-generated summaries, the average interpretability confidence score reaches 4.16, making the results accessible and reliable even for non-expert users. Overall, the framework reduces technical barriers, optimizes the use of remote sensing imagery, and provides a scalable solution for intelligent water quality monitoring and complex ecological applications.

1. Introduction

Water resources are closely related to human health and sustainable socioeconomic development (Chen et al., 2024). Groundwater quality assessment studies have highlighted the spatial variability of water quality and the influence of both natural geochemical processes and anthropogenic activities on water suitability for drinking (Jodhani et al., 2025a). However, industrialization and rapid urbanization have exacerbated surface water pollution, highlighting the urgent need for effective water quality monitoring (Ge et al., 2024; Zheng et al., 2024). Traditional monitoring methods, while accurate, are costly and time-consuming. Automated detection stations offer an alternative, but their high installation and maintenance costs limit spatial coverage. In contrast, remote sensing

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https://doi.org/10.1016/j.rsase.2026.101951
Received 12 October 2025; Received in revised form 1 February 2026; Accepted 24 February 2026
Available online 26 February 2026
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technology, with its advantages in large-scale observation and periodic monitoring, has become an indispensable tool for water quality assessment (Sun et al., 2024). Cloud-based platforms such as Google Earth Engine (GEE) provide scalable computational capabilities for environmental monitoring and analysis (Jodhani et al., 2025b).

Advances in satellite sensor technology have greatly increased the availability of remote sensing data, driving the rapid expansion of water quality monitoring applications. Satellite datasets from Landsat, Sentinel, and GF platforms are widely used for large-scale inversion, while UAV-based multispectral imagery supports small- and medium-scale monitoring (Wang et al., 2025). However, the sharp growth in data availability has also intensified the challenge of extracting meaningful insights from massive datasets (Chen et al., 2025). A major barrier is that applying diverse remote sensing imagery to water quality inversion requires substantial domain knowledge and advanced technical expertise (Jiang et al., 2025). Although cloud platforms such as Google Earth Engine (GEE) offer powerful computational capabilities (Khan et al., 2021; Li et al., 2025), current GEE-based workflows remain complex and heavily reliant on specialized knowledge and programming skills, which significantly limit accessibility for non-expert users (Hou et al., 2025c; Kao et al., 2025).

Recent advances in artificial intelligence (AI), particularly in natural language processing, have created new opportunities for the intelligent transformation of remote sensing applications (Singh et al., 2024). The powerful capabilities of large language models (LLMs) have introduced a new paradigm for intuitive human-computer interaction (Liu, 2025). Notably, several pioneering frameworks—including Remote Sensing ChatGPT (Guo et al., 2024), Chain-of-Programming (Hou et al., 2025a), AutoGEEval (Wu et al., 2025), GeoTools-GPT, GeoCogent, and LLM-Geo—have shown promising progress in building end-to-end workflows that translate natural language into geospatial analysis(Wei et al., 2025; Hou et al., 2025b; Wang et al., 2024). Collectively, these systems represent significant steps toward democratizing remote sensing technology through natural language interfaces.

However, when applied specifically to water quality monitoring, existing frameworks exhibit three critical limitations that constrain their practical utility. First, they lack domain-specific optimization for water quality parameters, which possess distinctive characteristics such as pronounced seasonal variability, dependence on specific spectral bands, and sensitivity to meteorological factors such as cloud cover. For example, estimating chlorophyll-a concentrations requires different temporal windows and spectral bands than monitoring total phosphorus or ammonia nitrogen. Second, current systems show limited contextual understanding of water quality semantics, particularly in interpreting time-sensitive queries (e.g., “spring fertilization period”) or managing trade-offs between cloud cover and temporal proximity in water bodies where parameters may change rapidly following precipitation events. Third, and most crucial for adoption, these frameworks offer minimal water quality-oriented semantic interpretation of results, leaving non-expert users unable to convert technical outputs into actionable environmental insights.

Against this backdrop, this study proposes a natural language-driven water quality inversion framework tailored to domain-specific challenges. Unlike general end-to-end remote sensing approaches, the proposed framework emphasizes optimization of each workflow component for water quality parameter retrieval. The main contributions are threefold:

First, we develop a water quality-optimized data selection mechanism that integrates domain-specific knowledge, including the seasonal temporal sensitivity of different parameters, spectral band requirements for specific water quality indicators, and intelligent trade-offs for cloud coverage that account for the unique optical properties of water bodies. Unlike general-purpose geospatial LLM systems, this approach embeds hydrological expertise directly into the data selection process.

Second, we introduce a retrieval-augmented generation (RAG) architecture specifically tailored to water quality semantics. This framework bridges the gap between natural language queries and technical implementation by leveraging a knowledge base built from water quality monitoring literature, seasonal parameter variation patterns, and established inversion algorithms. By doing so, it overcomes a key limitation of existing systems: the inability to effectively interpret domain-specific terminology and temporal references commonly encountered in water quality monitoring requests.

Third, we implement a result interpretation module that translates technical inversion outputs into clear and actionable environmental insights. This marks a significant improvement over existing frameworks, which focus only on code generation and leave interpretation entirely to the user.

By focusing on water quality monitoring rather than building a general-purpose geospatial LLM, our framework achieves higher data selection accuracy (93% temporal matching), stronger semantic understanding for time-sensitive queries (91% recall), and enhanced accessibility for non-expert users. Its primary innovation is this domain-specific design, which allows users to retrieve professional-grade water quality information through simple natural language queries without needing expertise in remote sensing or GEE operations. This approach provides a practical pathway for democratizing remote sensing in environmental management.

2. Related work

Advances in AI have accelerated the development of LLMs and multimodal vision-language models, unlocking new possibilities for intelligent remote sensing data analysis (Wagner et al., 2024; Yuan et al., 2025). Yet, remote sensing imagery differs fundamentally from natural images, creating unique challenges (Shu et al., 2025; Zhang et al., 2024). Unlike standard RGB images, remote sensing data often include multi-band, multidimensional features that require careful integration of geospatial, temporal, and environmental information. Additionally, the wide range of applications—spanning classification, segmentation, object detection, change detection, and parameter inversion—exceeds the generalization capacity of conventional models (Ge et al., 2025; Sun et al., 2024; Wen et al., 2025; Yuan et al., 2024).

To overcome these limitations, researchers have focused on developing foundation models specifically designed for remote sensing. SkyEyeGPT introduced a multimodal remote sensing instruction dataset (SkyEye-968k) to enable unified task processing (Zhan et al., 2025). EarthGPT-X employed cross-domain fusion training to handle heterogeneous data, incorporating a pixel-aware module for joint

reasoning across multimodal information (Zhang et al., 2025). RSGPT advanced benchmark-driven development through high-quality remote sensing image captioning (RSICap) datasets and standardized evaluation frameworks (Hu et al., 2025). Collectively, these studies highlight efforts to enhance models' comprehension of remote sensing semantics through large-scale datasets.

Similarly, several studies have examined the use of large remote sensing models in specialized applications. To address the requirements of specific subfields, researchers have explored methods that incorporate domain knowledge directly into model architectures (Liu et al., 2024). For example, Tree-GPT combines image understanding, domain expertise, and toolchains to create end-to-end workflows for forest monitoring (Du et al., 2023), while VectorLLM enables building footprint extraction through vertex-by-vertex regression (Zhang et al., 2025). Although these models have achieved significant progress, their applicability remains limited due to task specificity and dependence on specialized knowledge (Akinboyewa et al., 2024; Huanyu et al., 2025).

Beyond enhancing model capabilities, recent research has focused on leveraging large models and agent technologies to improve human-machine interaction in remote sensing. Systems such as Remote Sensing ChatGPT (Guo et al., 2024), AutoGEEval (Wu et al., 2025), Chain-of-Programming (Hou et al., 2025a), GeoTools-GPT, GeoCogent, and LLM-Geo have advanced the construction of end-to-end workflows, spanning natural language understanding to geospatial code generation(Wei et al., 2025; Hou et al., 2025b; Wang et al., 2024). Chat-RSC facilitates interactive remote sensing classification using state machines and agents (Pan et al., 2025), while GeoLLM-Squad employs multi-agent collaboration to tackle complex analytical tasks (Lee et al., 2025).

Despite advances in end-to-end remote sensing workflows, existing systems remain poorly suited for water quality monitoring.

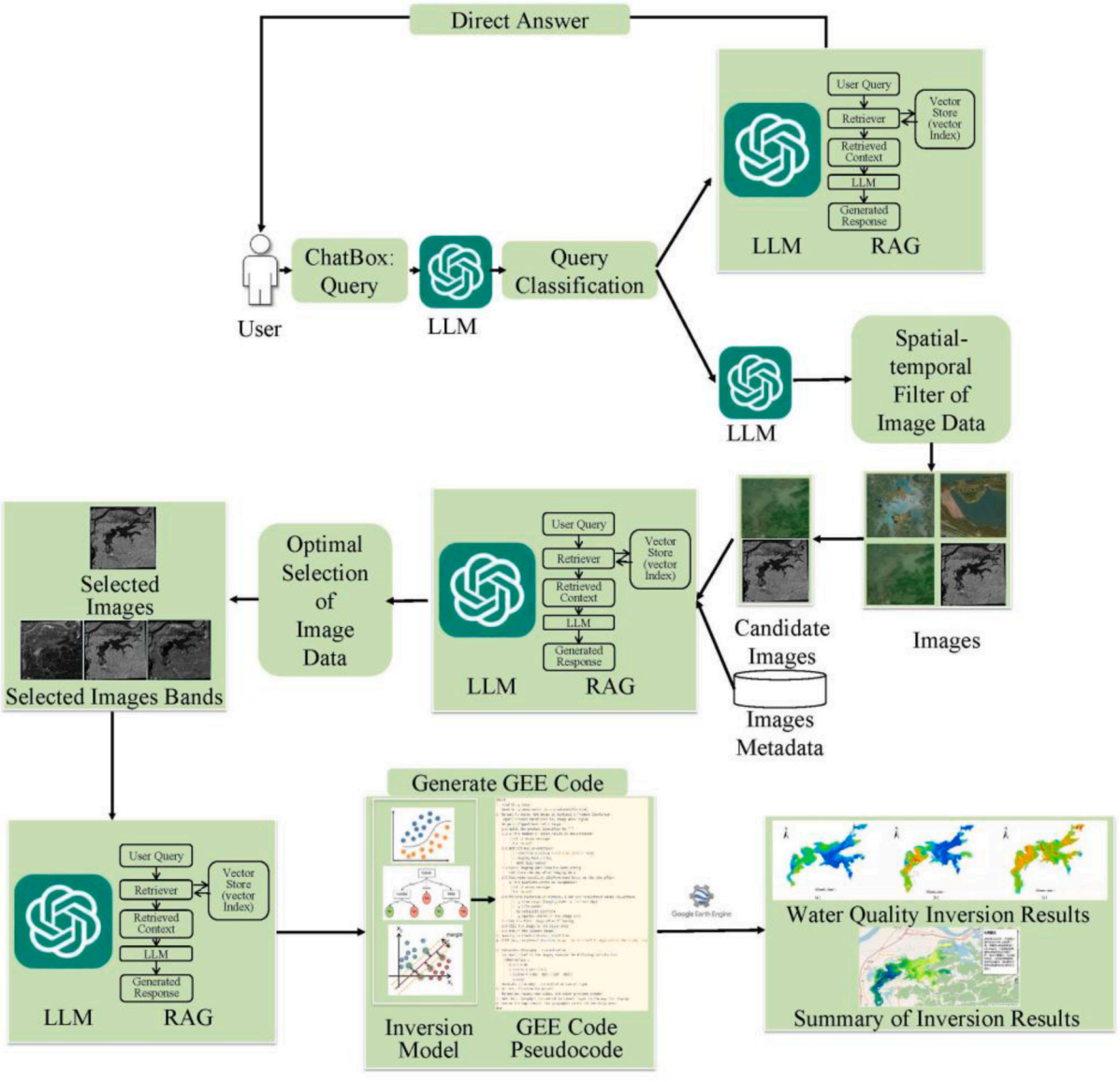


Fig. 1. Overall framework of the proposed method.

Major limitations include inadequate optimization for water quality inversion, limited semantic understanding of natural language queries—especially those with implicit temporal and spatial information—and low usability and interpretability for non-expert users. To address these gaps, this study proposes a natural language-driven framework specifically designed for water quality inversion, incorporating spatiotemporal and seasonal filtering, cloud coverage optimization based on parameter sensitivity, and spectral suitability evaluation. By integrating domain knowledge through RAG, the framework effectively bridges user intent and technical implementation while delivering interpretable results, thereby improving accessibility and decision support for a wide range of stakeholders.

3. Methodology

3.1. Overall structure of the proposed method

This study presents a water quality retrieval framework that integrates LLMs with GEE (Fig. 1), enabling end-to-end automation from user requirement interpretation to water quality retrieval. By combining LLMs, RAG techniques, and GEE-based image processing, the framework seamlessly incorporates critical functions—including data screening, feature selection, model selection, model retrieval, and result interpretation—thereby enhancing the efficiency and accuracy of water quality inversion.

(1) Requirement Interpretation and Task Classification Module

The system accepts user input in the form of natural language queries or graphical interaction commands (e.g., specifying spatial ranges and time intervals). It parses these inputs to determine the type of request. Routine water quality queries can be addressed directly by the LLM, whereas more specialized queries are handled through a combination of LLM and RAG. If the query involves water quality retrieval, it enters the retrieval workflow, triggering the spatiotemporal data screening module.

(2) Spatiotemporal Data Screening Module

Once a user request is identified as involving water quality retrieval, the system activates this module for data screening. It handles large-scale, multi-source, multi-resolution, multispectral, and multi-temporal datasets from platforms such as Landsat, Sentinel, GF, and UAVs. Selecting the most suitable dataset to match the user's request is critical for ensuring the accuracy and reliability of subsequent water quality retrieval.

Data screening in this module is a multi-step process that emphasizes spatiotemporal filtering of image data. Specifically, it retrieves candidate datasets from multiple sources that meet predefined spatial and temporal constraints. For spatial filtering, the system uses Turf.js in combination with the geometric features of each image. For temporal filtering, it leverages the semantic analysis capabilities of LLMs to interpret user-specified dates or time intervals and select images that satisfy the temporal requirements.

(3) Data Prioritization Based on Image Metadata Module

After spatiotemporal screening, the candidate datasets satisfy the user requirements; however, factors such as spectral properties, cloud coverage, spatial resolution, and alignment with retrieval targets can impact the results. Therefore, further prioritization of remote sensing data based on metadata is necessary. This step demands substantial domain knowledge and technical expertise and is crucial for accurately fulfilling the user's intent. Implementation details and a comparative evaluation of different approaches are presented later.

(4) Model Invocation and Retrieval Execution Module

This stage mainly utilizes the code generation capabilities of LLMs. First, based on the outputs from previous steps, the user's retrieval targets, and the prioritized image data, the system selects the appropriate model for water quality retrieval. Second, following GEE's runtime environment and coding standards, the LLM generates retrieval code that can be executed within GEE. Finally, the system executes the GEE retrieval code, obtains the retrieval results, and performs the necessary output operations.

(5) Result Semantic Interpretation Module

In water quality inversion, to help non-specialists better understand remote sensing-derived results and accurately interpret water quality conditions, the system combines layered rendering of water quality parameters with the use of LLMs to summarize inversion outcomes. This approach improves both the readability and interpretability of water quality data.

For example, when retrieving lake water quality based on chlorophyll-a concentration and using the TSL trophic state index, the trophic state proportions are as follows: oligotrophic (<30) – 1.78%, mesotrophic (30–50) – 1.89%, slightly eutrophic (50–60) – 46.33%, moderately eutrophic (60–70) – 49.86%, and heavily eutrophic (>70) – 0.14%. Based on these values, describe the current water quality status of the lake.

LLM Summary: Based on the inversion results of chlorophyll-a concentration in the lake, the water bodies are predominantly eutrophic. Oligotrophic and mesotrophic waters account for only 1.78% and 1.89%, respectively, indicating that low-trophic-level

areas are minimal. Slightly eutrophic waters make up 46.33%, and moderately eutrophic waters 49.86%, together representing the vast majority and indicating that most of the lake is in a eutrophic state. Severely eutrophic waters account for only 0.14%, a negligible proportion. Overall, the lake is mainly characterized by slight to moderate eutrophication, with the proportion of severely eutrophic waters being almost negligible and the proportions of other trophic levels remaining relatively small.

3.2. Sample collection and test data

This study constructs a comprehensive, multi-source remote sensing dataset for natural language-driven water quality monitoring, integrating Sentinel-1/2, Landsat 8/9, GF-1/6, and UAV imagery with complete metadata, including image name, acquisition time, product identifier, cloud coverage percentage, and spatial footprint geometry. Unlike previous evaluations that relied on pre-filtered candidate sets, our dataset explicitly retains the full spatial context, enabling assessment of the model's true geospatial reasoning capabilities.

Each image in our dataset is annotated with its precise spatial footprint using GeoJSON polygon geometries, enabling direct assessment of spatial overlap between user queries and candidate imagery. The dataset comprises 2850 images covering diverse water bodies across China, with spatial extents ranging from small ponds of 0.1 km^2 to large lakes exceeding 500 km^2 and varying geographic complexity, from regular shapes to highly irregular coastlines.

To evaluate semantic reasoning in data selection, we designed queries with varying spatial complexity:

(1) Precise spatial references (e.g., “Shengjin Lake,” “30.5°N 117.2°E”)
(2) Relative spatial descriptions (e.g., “the upstream section of Chaohu Lake,” “within 5 km of the estuary”)
(3) Administrative boundaries (e.g., “water bodies in Anhui province,” “urban rivers of Hefei city”)
(4) Fuzzy spatial concepts (e.g., “the eastern part of Taihu Lake,” “surrounding waters of the island”)

Our evaluation framework incorporates a spatial reasoning module that processes user queries without pre-filtering candidate images. Instead, the system identifies suitable imagery from the full set of 2850 images based solely on the natural language description. Spatial matching is rigorously assessed using intersection-over-union (IoU) metrics between the user-specified region and image footprints, with IoU $\geq$0.8 considered a successful match.

Spatial matching is evaluated using the Intersection-over-Union (IoU) metric between the user-specified query region and the spatial footprint of each candidate image, with IoU $\geq$0.8 considered a successful match. This relatively strict threshold reflects the high sensitivity of water quality inversion to spatial completeness, as partial coverage of target water bodies may introduce mixed pixels or omit critical regions. Compared with lower IoU thresholds commonly used in general vision tasks, an IoU of 0.8 ensures near-complete spatial coverage while allowing limited geometric discrepancies caused by sensor characteristics and complex water body boundaries. This criterion provides a balanced and robust basis for assessing spatial reasoning performance.

To create representative samples, we generated 800 high-quality query-image pairs covering spatial, temporal, cloud coverage, and spectral band dimensions. Markdown is used as the storage format for sample data, with strict formatting standards established to ensure data readability. The dataset is split into a training set (n = 700) and a test set (n = 100) at a 7:1 ratio. The training set serves as the knowledge base for RAG strategies, while the test set is used to evaluate model performance on previously unseen queries that require genuine spatial reasoning.

For clarity, each query-image pair is represented in a tabular format with the following structure: (see Table 1).

This comprehensive dataset design goes beyond limited candidate sets, enabling a realistic evaluation of LLM spatial reasoning capabilities in water quality monitoring. It addresses the critical need for robust geospatial reasoning in natural language-driven remote sensing applications.

3.3. Data selection and optimization approach

Data optimization focuses on refining the selection of remote sensing imagery for inversion. As a key component of the overall framework, it requires careful consideration of factors such as spectral characteristics, cloud coverage, and the appropriateness of acquisition timing.

(1) Spectral Band Suitability Analysis

The composition of spectral bands in remote sensing imagery is crucial for the accuracy of water quality parameter inversion. Different parameters require data sources that provide access to specific spectral bands. For example, chlorophyll-a concentration is strongly correlated with Sentinel-2 bands B4 (Red, 665 nm) and B5 (Near-Infrared, 865 nm) ($R^2 > 0.75$) (Shi et al., 2022). In contrast,

Table 1
Schematic illustration of markdown table structure.

Index	Data Source	Product Identifier	Date	Cloud %	Question	Answer	Reason
1	Sentinel-2	S2A_XXXX	X	X	Example	Sentinel-2	Example
2	Landsat9	LC09_XXXX	X	X	Example	Landsat9	Example

the inversion of total phosphorus and total nitrogen depends on shortwave infrared bands (e.g., Sentinel-2 B11, 1610 nm) to account for interference from suspended matter (Yang et al., 2024). Sentinel-1, which lacks visible bands, is therefore unsuitable for directly inverting these parameters.

(2) Determination of Reasonable Temporal Intervals

Water quality parameters exhibit significant seasonal variability, requiring differentiated temporal constraint standards (Table 2). These temporal windows are not set arbitrarily but are based on typical seasonal patterns of water quality parameters (Staehr et al., 2023) and empirical knowledge of the study area, which informed the experimental parameters of this research. For example, during spring (the fertilization period), total phosphorus concentrations are less affected by rainfall, so imagery within a 7-day window is considered representative (e.g., an image acquired on April 1 reflects conditions on March 30). In contrast, the heavy rainfall of summer requires a tighter 3-day window to capture short-term fluctuations (e.g., an image from July 10 represents conditions on July 8). Chlorophyll-a concentrations remain relatively stable in winter, allowing a relaxed 14-day window, whereas ammonia nitrogen, highly sensitive to microbial decomposition, requires a stricter 5-day window in autumn.

(3) Cloud Coverage Considerations in Multi-Factor Trade-Offs

Cloud coverage has a critical impact on inversion accuracy, but must be considered alongside temporal constraints. While lower cloud coverage generally improves accuracy, selecting the image with the absolute lowest coverage is not always optimal, particularly during seasons with rapid water quality changes, such as summer. When multiple images have similarly low cloud coverage, priority is given to those closest to the target monitoring date. For example, images with cloud coverage below 20% are preferred. If coverage levels are comparable (e.g., 15% vs. 18%), total phosphorus inversion in summer should prioritize the image nearest the monitoring date (e.g., July 5 over July 8) to reduce deviations caused by short-term rainfall.

(4) Semantic Reasoning Strategy Comparison Design

To evaluate the effectiveness of different strategies for remote sensing data selection, this study designs three comparative experiments: Zero-shot strategy: The model is provided with no example inputs and generates responses solely based on the task description. This setup assesses the model's zero-shot generalization capability in data optimization. Few-shot strategy: A small set of manually curated examples is used to construct prompt templates, guiding the model to apply data optimization principles consistently across different scenarios. RAG strategy: Sample knowledge is transformed into semantic vectors using embedding methods, enhancing the model's ability to incorporate relevant background information. In this study, 700 samples are encoded into 768-dimensional embeddings using the BCEmbedding model (netease-youdao/bce-embedding-base_v1), forming a semantic knowledge base (NetEase Youdao, 2023). Within the Qwen2.5-7B-Instruct model, the top six most relevant entries are retrieved through semantic similarity matching to support the generation of optimized outputs.

3.4. GEE-based code generation

This study uses LLM-generated executable code on the GEE cloud platform to perform the final inversion of water quality parameters.

(1) Model and Tool Selection

A modular LLM architecture is adopted in which Qwen2.5-7B-Instruct handles lightweight tasks such as requirement parsing, task classification, and intent recognition to ensure efficiency. Qwen2.5-Coder-32B-Instruct serves as the core engine for complex algorithm construction and code generation within the GEE environment, leveraging its advanced reasoning and programming capabilities. Additionally, Qwen3-Embedding-8B is integrated for semantic vectorization to enhance domain adaptability and cross-module semantic understanding, enabling both computational efficiency and high-quality outputs within a unified framework.

(2) Construction of the Domain Knowledge Base

To enhance code generation accuracy, this study develops two types of domain knowledge bases: (i) a structured sensor feature

Table 2
Reasonable temporal ranges (days) of remote sensing imagery for different water quality parameters across seasons.

Water Quality Parameter/Season	Chlorophyll-a Concentration	Total Phosphorus	Total Nitrogen	Ammonia Nitrogen
Spring	7	7	7	5
Summer	3	3	3	3
Autumn	7	5	5	5
Winter	14	14	14	14

knowledge base, constructed from a review of spectral characteristics, preprocessing workflows, and metadata standards for multi-source remote sensing data, including Landsat 8/9, Sentinel-1/2, UAV, and GF-1/6; and (ii) a water quality inversion knowledge base, covers parameters such as chlorophyll-a, total phosphorus, and total nitrogen, compiling literature-validated inversion formulas, optimal parameter sets, and band combinations. These knowledge elements are then converted into semantic vectors using the embedding model, enabling precise, context-aware code generation.

(3) Code Generation

Code generation in this study proceeds in three steps. First, the LLM identifies the inversion target (e.g., chlorophyll-a or total nitrogen). Afterward, it retrieves the corresponding GEE code templates to serve as exemplars for generation. Finally, the model produces and reflectively evaluates the code to ensure alignment with domain knowledge and platform specifications. Failures are quantitatively categorized as structural omissions (e.g., missing key bands or preprocessing steps), syntax errors (e.g., API mismatches or misspellings), and logical conflicts (e.g., incorrect algorithm implementation). This workflow effectively integrates domain expertise with LLM capabilities, ensuring GEE-specific syntax compliance while enhancing adaptability to multi-source remote sensing data, thereby enabling a fully end-to-end inversion process from user request to final results.

3.5. Evaluation methodology

To comprehensively evaluate LLM performance in water quality retrieval tasks, this study establishes a multidimensional evaluation framework across five perspectives: (1) evaluation of user requirement analysis, (2) evaluation of the accuracy of water quality remote sensing data selection, (3) evaluation of water quality semantic task comprehension, (4) evaluation of execution and output of water quality retrieval tasks, and (5) evaluation of water quality retrieval results. Specifically, these are detailed as follows:

3.5.1. Evaluation of user requirement analysis

To assess the ability of quantitative models to interpret user requirements and the reliability of task classification, this study introduces the task parsing consistency (TPC) metric. The TPC is calculated using the following formula:

$$\mathrm{TPC} = \frac{\mathrm{N_{correct}}}{\mathrm{N_{total}}} \tag{1}$$

Here, $\mathrm{N_{correct}}$ denotes the number of task types correctly identified by the model, and $\mathrm{N_{total}}$ represents the total number of task samples.

This metric quantifies the model's task comprehension ability, ensuring that subsequent modules operate based on accurate task interpretation and thereby maintaining the reliability and traceability of the entire process.

3.5.2. Evaluation of the accuracy of water quality remote sensing data selection

This study uses two metrics, namely “temporal matching rate” and “spatial matching rate,” to assess the model's performance in remote sensing image selection. The temporal matching rate measures the proportion of selected images that fall within the user-specified time range, while the spatial matching rate quantifies the proportion of images that adequately cover the target area.

$$\text{Temporal Matching Rate} = \frac{\mathrm{N_t}}{\mathrm{N}} \times 100\% \tag{2}$$

$$\text{Spatial Matching Rate} = \frac{\mathrm{N_s}}{\mathrm{N}} \times 100\% \tag{3}$$

Here, N denotes the total number of test queries, each associated with one model-selected image; N_t denotes the number of images whose acquisition times satisfy the user-specified temporal requirements; and N_s represents the number of images that completely cover the target area.

3.5.3. Evaluation of water quality semantic task comprehension

To systematically assess the model's performance in semantic parsing and classification, this study employs both classification metrics and ranking-based metrics for quantitative evaluation. In addition, an in-depth analysis is conducted to examine the model's response effectiveness across different question types:

3.5.3.1. Ranking evaluation metrics. To strengthen the statistical rigor of the data selection module evaluation, this study introduces metrics tailored for ranking scenarios, including mean reciprocal rank (MRR) and top-k accuracy. The methodology for constructing ranking evaluation units and their corresponding ground truth is also specified.

(1) Definition of Evaluation Units for Ranking Tasks

In the water quality inversion data selection process, the model must choose the images that best satisfy user requirements from multiple candidates. Accordingly, this study defines “each user query” as an evaluation unit, treating the corresponding candidate

image list as a separate ranking task. The ranking sequence generated by the model is then compared with the ground-truth optimal ranking, constructed by experts, to evaluate ranking quality and the model's selection capability.

(2) Construction Method for Ground-Truth Ranking

To ensure rigorous evaluation, ranking rules are established based on the core data quality requirements for remote sensing water quality inversion, covering four key categories:

Temporal: According to the "parameter-seasonal time thresholds" in Table 2, images closer to the target date are assigned higher scores, while images outside the acceptable time range are downgraded.

Spatial: Images that fully cover the user-specified region are given the highest priority, whereas images with only partial coverage receive lower priority.

Cloud Cover: Images with cloud coverage below 20% are preferred. When cloud coverage is similar, priority is given to images closer to the target date.

Spectral Bands: Only images containing the spectral bands required for the target water quality parameters receive full scores.

Ultimately, each query is associated with a unique "optimal ranking," which serves as the ground truth for that query.

(3) Ranking Metrics

This study employs MRR and top-k accuracy to quantify the model's performance in ranking tasks with multiple candidate images:

$$\mathrm{MRR} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{\mathrm{rank}_i} \tag{4}$$

$$\text{Top-k accuracy} = \frac{C_k}{N} \tag{5}$$

Here, $rank_i$ denotes the position of the ground-truth optimal image in the model-generated ranking.

C_k indicates the number of queries for which the ground-truth optimal image is ranked within the top-k positions. A higher MRR corresponds to the optimal image appearing earlier in the ranking. Top-k accuracy measures whether the ground-truth optimal image appears within the top k positions. In this study, top-1 and top-3 accuracy are reported to comprehensively assess the model's ability to identify high-priority images.

3.5.3.2. Classification evaluation metrics. The model's semantic parsing ability in classification tasks is systematically evaluated by comparing true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), followed by calculation of accuracy, precision, recall, and F1-score. The formulas for these metrics are as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \tag{6}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{7}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{8}$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{(\text{Precision} + \text{Recall})} \tag{9}$$

where TP represents the number of true positives, TN the number of true negatives, FP the number of false positives, and FN the number of false negatives.

3.5.3.3. Error analysis of question types and reasoning strategies. To further assess the LLM's performance across different question types, test set samples are categorized into three groups: "temporal," "cloud coverage," and "band-related." The distribution of errors under different reasoning strategies is analyzed to guide strategy selection and model optimization. Error combinations for the strategies are defined as follows: (a) zero-shot or few-shot incorrect, RAG incorrect; (b) zero-shot or few-shot incorrect, RAG correct; (c) zero-shot, few-shot, and RAG all correct; (d) zero-shot and few-shot correct, RAG incorrect.

3.5.4. Evaluation of execution and output of water quality retrieval tasks

This study uses the "code runnable rate" to evaluate the code generated by the model. The code runnable rate measures the ability of the model-generated code to execute successfully on the GEE platform. Its calculation formula is as follows:

$$\text{Code Runnable Rate} = \frac{\text{Number of Successfully Executable Codes}}{\text{Total Number of Generated Code Samples}} \times 100\% \tag{10}$$

3.5.5. Evaluation of water quality retrieval results

An inverted image is regarded as consistent with the output if its visual appearance aligns with the expected result, thus ensuring the objectivity and verifiability of the evaluation metric. To assess the reliability of the model during the result interpretation phase, the study employed the explanation reasonableness score (ERS), evaluated by experts or users using a standardized scoring rubric (Table 3).

ERS evaluates the explanation along three dimensions, including scientific validity, which reflects adherence to remote sensing principles; consistency, which assesses how well the interpretation aligns with the retrieval results; and clarity and completeness, which considers whether the key reasoning, limitations, and conclusions are clearly and comprehensively expressed. Experts scored each dimension on a 1–5 scale, as summarized below:

The ERS for a set of results is calculated as the average score across all experts:

$$\mathrm{ERS} = \frac{1}{\mathrm{N}} \sum_{\mathrm{i}=1}^{\mathrm{N}} \mathrm{S_i} \tag{11}$$

where S_i represents the score assigned by the *i*-th expert, N is the total number of experts.

4. Results and discussion

4.1. Task parsing consistency

To implement a procedural approach for intelligent remote sensing water quality question answering, the system's query parsing module must accurately interpret users' natural language descriptions and graphical interaction commands. This allows the system to distinguish between routine water quality inquiries, domain-specific questions, and complex tasks that require entry into the inversion process. Using Qwen2.5-7B-Instruct, we systematically evaluated TPC under three reasoning strategies: zero-shot, few-shot, and RAG. The results are shown in Table 4. Under the zero-shot strategy, without contextual or domain constraints, identification of temporal ranges, spatial regions, and inversion-related elements is prone to errors, yielding a TPC of 0.64. The few-shot strategy, by constraining the task semantic space with examples, enables more stable query type discrimination, increasing TPC to 0.80. In contrast, RAG, which integrates external domain knowledge retrieval, substantially improves the model's ability to consistently parse query intent, domain-specific elements, and task workflow (e.g., determining whether to proceed with spatiotemporal filtering and inversion), achieving a TPC of 0.92. These results highlight that incorporating professional knowledge is essential for enhancing semantic recognition and workflow routing in complex water quality tasks, providing a reliable foundation for the system's automated inversion execution.

4.2. Spatiotemporal data screening

The results are presented in Table 5. Spatial matching was performed by comparing user queries against the full set of 2850 images without any pre-filtering, using a sample of 100 queries for detailed evaluation. This approach achieved an accuracy of 94%, demonstrating the framework's ability to interpret a wide range of spatial descriptions—from precise coordinates to diffuse regional references. It also substantially outperformed the zero-shot (76%) and few-shot (81%) baselines, which were evaluated on the same sample set.

Temporal reasoning was systematically evaluated across zero-shot, few-shot, and RAG strategies, using a sample of 100 temporally annotated queries. The zero-shot strategy achieved 74%, reflecting limited capability for dynamic temporal inference, while few-shot learning provided a modest improvement (77%), dependent on the quality of examples. In contrast, the RAG strategy, which incorporates predefined season-parameter temporal rules, achieved 93% temporal matching, effectively resolving semantic ambiguities.

Collectively, these results demonstrate that a geospatially and temporally enhanced RAG architecture markedly improves both spatial and temporal understanding, outperforming conventional zero-shot and few-shot approaches in large-scale geospatial-temporal reasoning tasks. This aligns with recent findings by Yu et al. (2025), who proposed the Spatial-RAG framework. By incorporating sparse spatial retrieval, Spatial-RAG extends traditional RAG methods to spatial tasks, enhancing the model's ability to address real-world spatial reasoning challenges.

Table 3
Standardized scoring rubric.

Score	Scientific Validity	Consistency	Clarity and Completeness
5	Fully follows principles; no errors	Fully matches results; all key features reflected	Clear logic; complete conclusions; reasoning and limitations fully explained
4	Mostly follows principles; minor deviations	Largely consistent; main features reflected	Logic clear; conclusions mostly complete; reasoning and limitations mostly explained
3	Partially follows principles; some errors	Partially consistent; some key features missing	Logic acceptable; conclusions partial; reasoning or limitations unclear
2	Significant deviations; incorrect interpretations	Poor consistency; important features missing or misleading	Logic unclear; conclusions incomplete; reasoning or limitations insufficient
1	Major violations; largely incorrect	Severely inconsistent; results not reflected	Difficult to understand; conclusions missing or unclear

Table 4
Comparison of task parsing consistency (TPC) across three reasoning strategies.

Strategy	TPC
Zero-shot	0.64
Few-shot	0.80
RAG	0.92

Table 5
Comparison of spatial and temporal matching rates across three reasoning strategies (%).

Strategy	Spatial Matching Rate (%)	Temporal Matching Rate (%)
Zero-shot	76	74
Few-shot	81	77
RAG	94	93

4.3. Evaluation results for water quality semantic tasks

4.3.1. Evaluation of ranking tasks

To strengthen the statistical rigor of the data prioritization module evaluation, the ranking metrics MRR and top-k accuracy were calculated, with the results illustrated in Fig. 2. The RAG strategy demonstrates superior performance across key ranking indicators: MRR reaches 0.85, top-3 accuracy reaches 0.97, and top-1 accuracy improves to 0.73, highlighting a clear performance advantage. Compared with zero-shot and few-shot strategies, RAG consistently ranks the truly optimal images at the top, indicating that the retrieval-augmented mechanism effectively enhances the model's understanding of image prioritization rules. The moderate improvement observed for the few-shot strategy suggests that example-based learning can partially enhance ranking reliability, but its effect remains substantially weaker than the gains achieved through integration of external knowledge. Overall, the significant improvements of RAG across all ranking metrics demonstrate that the retrieval-augmented mechanism can effectively improve the quality of prioritized remote sensing image data.

4.3.2. Evaluation of classification tasks

To comprehensively assess the semantic understanding capability of the inversion framework during the data selection stage, this study employed four key metrics—accuracy, precision, recall, and F1 score—to perform a detailed analysis of three core question types: temporal, cloud coverage, and spectral bands (Table 6, Fig. 3).

(1) Accuracy

Under zero-shot conditions, the model achieved a screening accuracy of 0.49, indicating basic generalization but leaving substantial room for enhancement. Incorporating a small set of examples through few-shot learning improved accuracy to 0.52, showing that example-based guidance can modestly improve the model's discriminative ability. The addition of the RAG strategy further

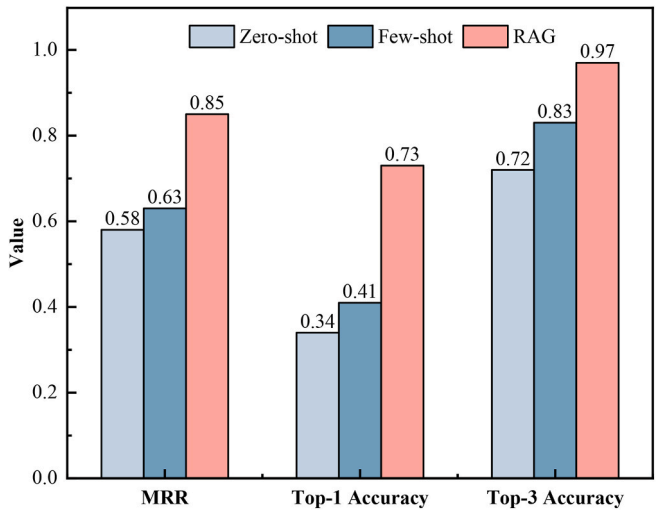


Fig. 2. Performance comparison of the three strategies on ranking tasks.

Table 6
Performance comparison of water quality remote sensing semantic classification across different question-answering strategies.

Evaluation Metrics	Zero-shot	Few-shot	RAG
TP	24	30	53
FP	20	19	23
TN	25	22	19
FN	31	29	5
Accuracy	0.49	0.52	0.72
Precision	0.55	0.61	0.70
Recall	0.44	0.51	0.91
F1-score	0.48	0.56	0.79

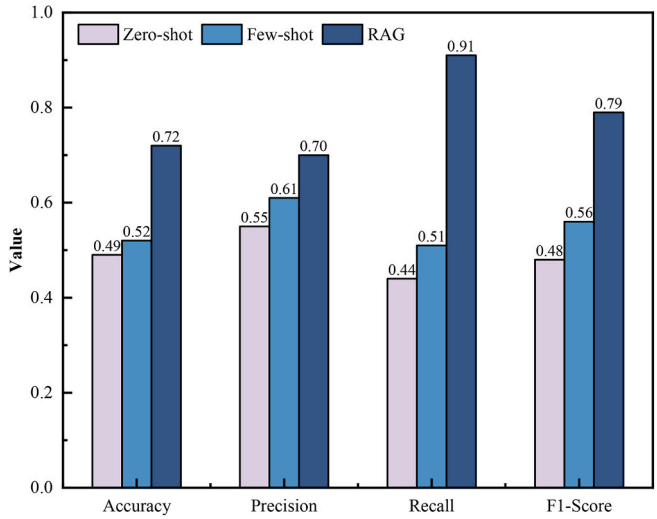


Fig. 3. Visual comparison of water quality remote sensing semantic classification performance across different question-answering strategies.

increased accuracy to 0.72, demonstrating that integrating external knowledge bases can markedly enhance data selection precision. These results emphasize the importance of data optimization in remote sensing image screening, especially for applications such as water quality inversion that rely on high-quality input data.

(2) Precision

Experiments showed that in the zero-shot stage, relying solely on the LLM's intrinsic knowledge and reasoning, precision reached 0.55, reflecting basic cognitive ability but limited effectiveness. In the few-shot stage, introducing a small set of examples increased precision to 0.61, indicating that the model began to apply screening rules such as spectral bands, cloud coverage, and temporal validity. However, owing to the limited number of examples, the model's judgments remained somewhat biased. In the RAG stage, by incorporating external knowledge retrieval, the model effectively integrated additional information, significantly enhancing task comprehension. Precision further increased to 0.70, demonstrating that retrieval-augmented knowledge strengthens semantic understanding and quantitatively validating the effectiveness of the RAG strategy in remote sensing image analysis.

(3) Recall

Recall, a key metric for evaluating the model's ability to identify positive samples is critical for the reliability of water quality monitoring. Experimental results indicate that under the RAG strategy, recall increased substantially from 0.44 in zero-shot mode to 0.91. This demonstrates that incorporating a semantic enhancement mechanism enables the AI agent to more comprehensively capture essential information from user requirements, such as temporal constraints and spectral band specifications. Recent reviews have also highlighted that RAG methods can significantly improve model recall when augmented with external knowledge, which aligns with the experimental findings of this study (Gao et al., 2023).

(4) F1-score

The F1-score was 0.48 in the zero-shot stage, increased to 0.56 in the few-shot stage with the inclusion of a small set of examples,

and ultimately reached 0.79 in the RAG stage, showing a clear trend of continuous improvement. This trend indicates that the model maintains prediction accuracy while achieving greater completeness in recall.

Compared with zero-shot and few-shot approaches, the RAG strategy demonstrates superior semantic understanding and task adaptability. However, practical applications still reveal certain limitations. For example, when prompts lack selectable images or contain only a single suitable image, the model may supplement answers from the knowledge base. Additionally, in cases of high cloud coverage, the model occasionally selects images with excessive clouds as useable data. For questions such as "whether water quality has improved," the model generally analyzes only the current state and cannot perform temporal logical reasoning. Moreover, when assessing whether an image contains the spectral bands required for water quality monitoring, the model's judgments can be inconsistent across different contexts, indicating that its understanding of remote sensing parameters still needs further refinement.

4.3.3. Analysis of error trends across question types and reasoning strategies

The response generated by RAG is shown in Fig. 4. According to Table 7 and Fig. 5, the RAG strategy demonstrates the strongest performance in handling temporal questions, successfully correcting 45% of samples that were misclassified under zero-shot or few-shot strategies. This result highlights RAG's robust capability for understanding temporal constraints. However, significant limitations remain in understanding remote sensing parameters, particularly for spectral band-related questions, where performance is weakest. In this category, 56% of samples were misclassified across all strategies, with RAG correcting only 19% of these errors. In addition, 6% of samples represented cases in which RAG failed while the other two strategies succeeded. This issue mainly stems from the model's limited ability to identify key spectral bands and its lack of domain-specific remote sensing knowledge. Even in cases where zero-shot or few-shot strategies correctly classified some samples, RAG could still produce errors.

4.4. Code generation and inversion

4.4.1. Code runnable rate

In this study, to evaluate the effectiveness of the proposed code generation and inversion framework, we compared it with three baseline models: KIMI-K1.5, Deepseek-V3, and Qwen2.5-Coder-32B-Instruct. Experimental results revealed differences in the performance of these models on the water quality inversion task, as illustrated in Fig. 6.

As illustrated in Fig. 6, the Qwen2.5-Coder-32B-Instruct + RAG model achieves a substantially higher code runnable rate than the other baseline models, reaching 87%, which represents a 36% improvement over the standalone Qwen2.5-Coder-32B-Instruct model (51%). This result indicates that integrating RAG technology can effectively enhance the model's comprehension of user requirements in water quality inversion tasks and significantly improve the quality of the generated code. In comparison, KIMI-K1.5 (47%) and Deepseek-V3 (61%) perform noticeably lower than the RAG-integrated model, suggesting that relying solely on conventional deep learning models for water quality inversion still has inherent limitations.

The GEE code snippets generated based on Qwen2.5-Coder-32B-Instruct and the RAG framework, along with the corresponding water quality retrieval results, are shown in Fig. 7.

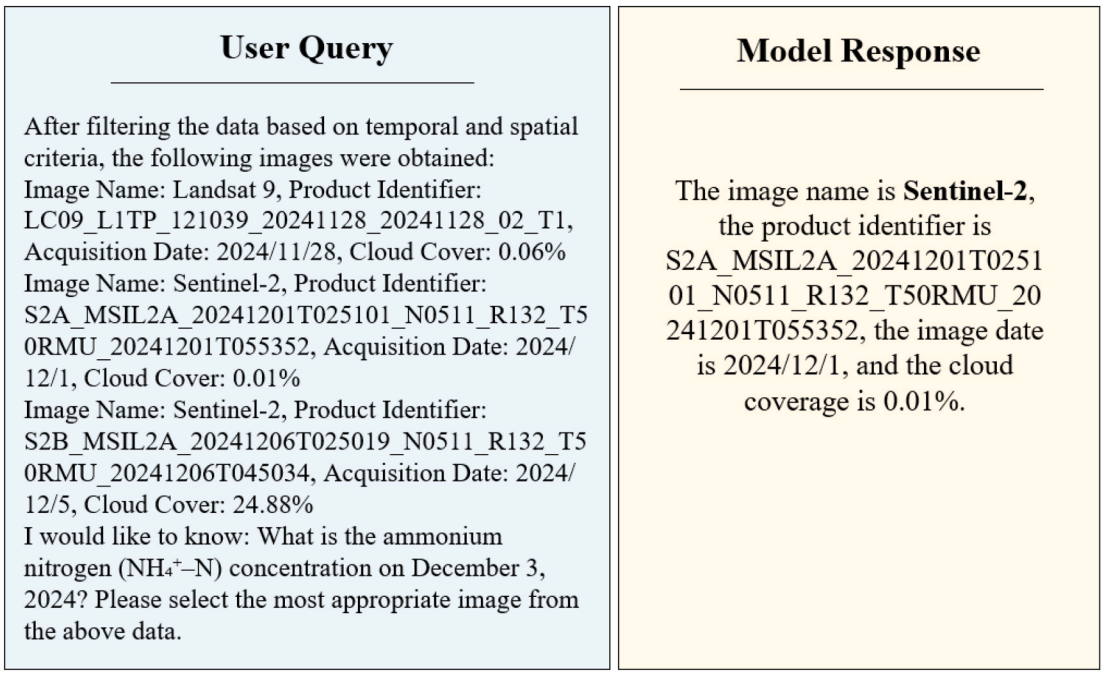


Fig. 4. Illustration of RAG response.

Table 7
Distribution of predicted combination proportions (%) for different types of water quality semantic questions.

Model Prediction Combination	Temporal Questions	Cloud-Cover Questions	Spectral Band Questions
Failure in both zero-shot/few-shot and RAG	12	27	56
Zero-shot/few-shot failure with RAG success	45	49	19
Success in zero-shot, few-shot, and RAG	37	24	19
Correct zero-shot/few-shot with RAG failure	6	0	6

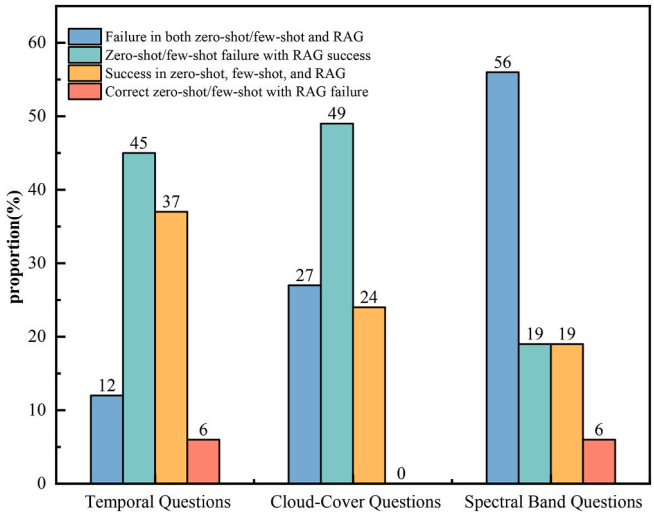


Fig. 5. Analysis of predicted combination percentages across different types of water quality semantic questions.

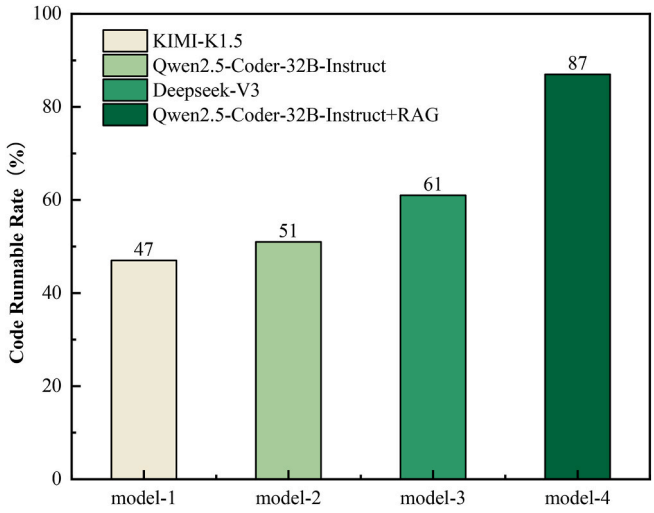


Fig. 6. Comparison of code runnable rates across different models.

4.4.2. Analysis of causes of code execution errors

Although the Qwen2.5-Coder-32B-Instruct + RAG framework significantly improves overall performance in the water quality inversion task, a certain proportion of erroneous code still appears in its outputs. To better understand these shortcomings, it is important to analyze the potential causes of these errors.

(1) Server-Side Computation Errors

During code generation, some errors stem from the handling of ee.Image server-side objects. Specifically, if an ee.Image object is incorrectly placed within an evaluate() callback function, mathematical operations on the results cannot be executed. While simple

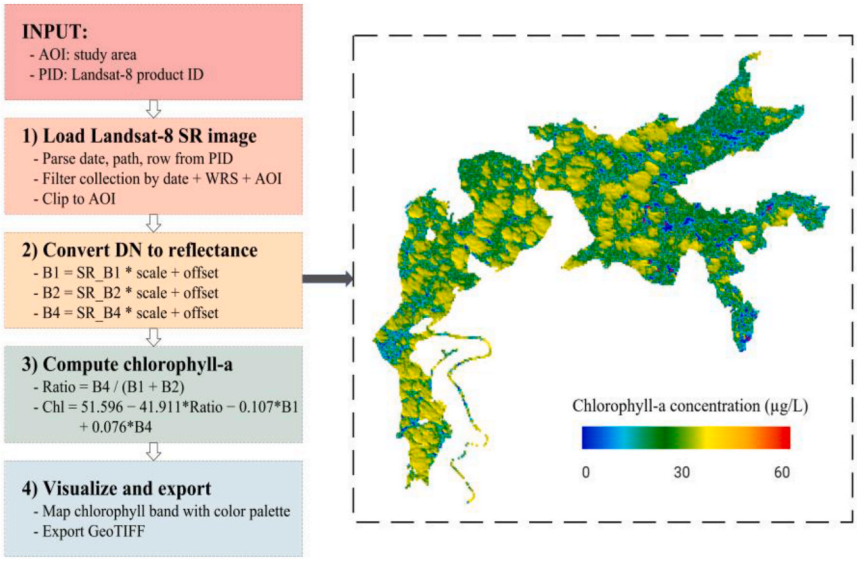


Fig. 7. Generated gee code and corresponding water quality retrieval outcomes.

operations, such as band ratio calculations, may succeed through implicit type conversion, more complex computations (e.g., exponentiation or polynomial regression) can fail owing to interruptions in the server-side computation graph. For example, using image.evaluate(function(img){ … }) prevents proper execution of server-side computations for ee.Image objects. To resolve this issue, all calculations on images must be performed on the server side rather than inside an evaluate() callback function.

(2) Band Definition Errors

In certain inversion tasks, spectral bands required by the inversion formulas were not explicitly defined, resulting in runtime errors. For example, when calculating total nitrogen concentration, the code referenced an undefined B3 band, even though only B1, B2, B4, and B6 bands were present in the dataset. To avoid such issues, the necessary bands must be explicitly defined in the code, for example: var B3 = image.select('B3');. Improper handling of band selection prevents correct execution and negatively impacts inversion results. Therefore, when generating code, it is essential to ensure that the definition and usage of each band are specified.

(3) Result Presentation Errors

Although some code executes successfully, the results are not properly displayed in certain cases. This is mainly due to missing map display calls (e.g., Map.addLayer())or the absence of appropriate visualization parameters (e.g., {min: 0, max: 1}), which prevents layers from being rendered correctly. Moreover, if specific areas of interest (AOIs) are not used to clip the resulting imagery, computations may extend beyond the target region, reducing visualization accuracy. To address these issues, code generation should include correct calls to map display interfaces, for example: Map.addLayer(image, {min: 0, max: 1}, 'water quality layer'), and specify appropriate visualization parameters to ensure proper layer rendering. Furthermore, when working with regional data, imagery should be accurately clipped using, for example, var clippedImage = image.clip(aoi);, so that the layer extent matches the target region and visualization precision is improved.

4.5. Interpretation of water quality inversion results and reliability analysis

In this study, water quality visualization was integrated with an LLM-based summarization module to optimize the interpretation of water quality inversion data. The results indicate that this module achieved an average ERS of 4.16, demonstrating its effectiveness in enhancing non-experts' understanding of remote sensing inversion results. This approach transforms complex numerical data into intuitive and easily comprehensible expressions, enabling users to quickly grasp the status of water quality in various water bodies. Further expert evaluation confirmed that the module performs well in terms of data interpretation accuracy, completeness of conclusions, and clarity of presentation, indicating that the interpreted results are both reliable and scientifically valid. Overall, the visualization combined with natural language summarization provides an effective means for disseminating and applying water quality monitoring results, offering important implications for improving the efficiency and accuracy of water environment management decision-making.

5. Conclusion

This study presents a natural language-driven framework for water quality inversion that integrates LLMs with GEE to address the complexity and domain knowledge demands of traditional remote sensing monitoring. The main findings are summarized as follows:

(1) Task Parsing Consistency: The RAG strategy achieves a TPC of 0.92, outperforming zero-shot (0.64) and few-shot (0.80), demonstrating that external domain knowledge significantly improves semantic understanding and workflow routing for complex queries.
(2) Spatiotemporal Data Screening: The framework achieves 94% spatial matching rate and 93% temporal matching rate, effectively interpreting both precise coordinates and vague regional or seasonal references, highlighting the value of domain-guided RAG for imagery selection.
(3) Water Quality Semantic Tasks: In classification and ranking, RAG improves performance, achieving an accuracy of 0.72, recall of 0.91, and top-3 ranking accuracy of 0.97, confirming the model's ability to capture key temporal and spectral constraints.
(4) Code Generation and Inversion: The RAG-enhanced Qwen2.5-Coder achieves a runnable code rate of 87%, significantly outperforming baseline models and reliably translating natural language queries into executable inversion workflows.
(5) Interpretation and Reliability: By integrating visualization with LLM-based summarization, the framework achieves an average ERS of 4.16, enhancing interpretability and reliability for non-expert users.

Overall, the framework reduces technical barriers, optimizes the use of accumulated imagery, and offers a scalable approach for intelligent water quality monitoring.

6. Limitations and Future Work.

While this study demonstrates significant advances in natural language-driven water quality inversion, several limitations should be acknowledged:

(1) Spectral band constraints: Performance is limited by sensor spectral capabilities; for example, the lack of red-edge bands in Landsat restricts inversion of certain parameters in turbid waters. Integration of hyperspectral data (e.g., PRISMA, EnMAP) may alleviate this issue.
(2) Temporal resolution trade-offs: Dependence on satellite revisit cycles may miss rapid water quality changes during extreme events. Integrating high-frequency in-situ or drone observations could improve temporal coverage.
(3) Regional transferability: Validation is currently focused on Chinese lakes with specific optical properties, and performance may vary across coastal waters, sediment-rich rivers, or different climatic zones. Broader global validation and region-specific algorithms are needed.
(4) Semantic ambiguity handling: Although RAG improves semantic understanding, highly ambiguous queries remain challenging. Interactive clarification mechanisms could enhance robustness.

Future research directions include: (a) adaptive band selection aligned with available sensors; (b) hierarchical monitoring integrating satellite, aerial, and in-situ data; (c) global water type classification to improve transferability; and (d) dialog-based query refinement for ambiguous inputs.

CRediT authorship contribution statement

Ming Liang: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Luyao Zhang:** Writing – original draft, Visualization, Validation, Software, Investigation, Data curation. **Jun Yan:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **ShanShan Wang:** Writing – review & editing, Writing – original draft, Resources, Formal analysis, Data curation. **Chengchen Liu:** Visualization, Validation, Investigation, Data curation.

Code availability section

Name of the code/library: code Contact: e-mail: lming@mail.ccnu.edu.cn.
Program language: JavaScript.
The source codes are available for downloading at the link: https://github.com/tonymm88/code.

Ethical statement

This study is based exclusively on publicly available remote sensing data and computational methodologies. No human participants, animal subjects, or personally identifiable information were involved in this research. All satellite imagery and environmental datasets used in this work were sourced from open-access platforms, including Google Earth Engine and other publicly archived repositories, in full compliance with their respective data usage policies. The proposed framework leverages large language models to enhance accessibility and usability of remote sensing tools, without generating or disseminating harmful, biased, or unethical content. The authors declare no conflicts of interest and confirm that this manuscript is original, has not been published previously, and is not

under consideration for publication elsewhere.

Funding

This research was supported by the Key Research and Development Project of Anhui Province (grant number 2022l07020027), the Natural Science Foundation of Anhui Province (grant number 1908085QD164), and the Natural Science Research Project of Colleges and Universities of Anhui Province (grant numbers 2022AH050095 and 2023AH053251).

Declaration of competing interest

The authors declare no competing interests. The authors have no financial or personal relationships that could inappropriately influence or bias the content of this work.

Data availability

Data will be made available on request.

References

Akinboyewa, T., Ning, H., Lessani, M.N., Li, Z., 2024. Automated floodwater depth estimation using large multimodal model for rapid flood mapping. Comput Urban Sci 4, 12. https://doi.org/10.1007/s43762-024-00123-3.

Chen, J., Wang, K., Li, X., Zhao, X., Cheng, X., Liu, Z., Zhang, J., Zhang, S., 2024. Remote sensing inversion of water quality parameters (TSM, Chl-a, and CDOM) in subtidal seaweed beds and surrounding waters. Ecol. Indic. 167, 112716. https://doi.org/10.1016/j.ecolind.2024.112716.

Chen, L., Liu, L., Liu, S., Shi, Z., Shi, C., 2025. The application of remote sensing technology in inland water quality monitoring and water environment science: recent progress and perspectives. Remote Sens. 17, 667. https://doi.org/10.3390/rs17040667.

Du, S., Tang, S., Wang, W., Li, X., Guo, R., 2023. Tree-GPT: modular large language model expert system for forest remote sensing image understanding and interactive analysis. https://doi.org/10.48550/arXiv.2310.04698.

Ge, J., Zhang, X., Zheng, Y., Guo, K., Liang, J., 2025. RSTeller: scaling up visual language modeling in remote sensing with rich linguistic semantics from openly available data and large language models. ISPRS-J. Photogramm. Remote Sens. 226, 146–163. https://doi.org/10.1016/j.isprsjprs.2025.05.002.

Ge, Y., Shen, F., Sklenička, P., Vymazal, J., Baxa, M., Chen, Z., 2024. Machine learning for Cyanobacteria inversion via remote sensing and AlgaeTorch in the Třeboň fishponds, Czech Republic. Sci. Total Environ. 947, 174504. https://doi.org/10.1016/j.scitotenv.2024.174504.

Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, M., Wang, H., 2023. Retrieval-augmented generation for large language models: a survey. https://doi.org/10.48550/arXiv.2312.10997.

Guo, H., Su, X., Wu, C., Du, B., Zhang, L., Li, D., 2024. Remote sensing ChatGPT: solving remote sensing tasks with ChatGPT and visual models. IGARSS 2024, 11474–11478. https://doi.org/10.1109/IGARSS53475.2024.10640736.

Hou, S., Jiao, H., Shen, Z., Liang, J., Zhao, A., Zhang, X., Wang, J., Wu, H., 2025a. Chain-of-programming (CoP): empowering large language models for geospatial code generation task. Int. J. Digit. Earth 18, 2509812. https://doi.org/10.1080/17538947.2025.2509812.

Hou, S., Jiao, H., Liang, J., Shen, Z., Zhao, A., Wu, H., 2025b. GeoCogent: an LLM-based agent for geospatial code generation. Int. J. Geogr. Inf. Sci. 0 (0), 1–34. https://doi.org/10.1080/13658816.2025.2549460.

Hou, S., Shen, Z., Zhao, A., Liang, J., Gui, Z., Guan, X., Li, R., Wu, H., 2025c. GeoCode-GPT: a large language model for geospatial code generation. Int. J. Appl. Earth Obs. Geoinf. 138, 104456. https://doi.org/10.1016/j.jag.2025.104456.

Hu, Y., Yuan, J., Wen, C., Lu, X., Liu, Y., Li, X., 2025. RSGPT: a remote sensing vision language model and benchmark. ISPRS-J. Photogramm. Remote Sens. 224, 272–286. https://doi.org/10.1016/j.isprsjprs.2025.03.028.

Huanyu, C., Yingcheng, G., Mengting, X., Qiuyuan, Z., Liu, W., 2025. A few-shot collapsed transmission Tower detection method combining large and small models in remote sensing image. IEEE Access 13, 41670–41681. https://doi.org/10.1109/ACCESS.2025.3547831.

Jiang, W., Ding, X., Kong, F., Luo, G., Long, T., Pang, Z., Cui, S., Liu, J., Adam, E., 2025. How hydrological extremes affect the Chlorophyll-a concentration in inland water in Jiujiang City, China: evidence from satellite remote sensing. ISPRS Int. J. GeoInf. 14, 85. https://doi.org/10.3390/ijgi14020085.

Jodhani, K.H., Gupta, N., Dadia, S., Patel, H., Patel, D., Jamjareegulgarn, P., Singh, S.K., Rathnayake, U., 2025a. Sustainable groundwater management through water quality index and geochemical insights in Valsad, India. Sci. Rep. 15, 8769. https://doi.org/10.1038/s41598-025-92053-1.

Jodhani, K.H., Sachapara, N.A., Patel, M., Gupta, N., Patel, D.P., Singh, S.K., Rathnayake, U., 2025b. Flood and erosion assessment of the Sabarmati River Basin: integrating big data in RUSLE and Google Earth Engine. Environ. Earth Sci. 84, 22. https://doi.org/10.1007/s12665-025-12676-5.

Kao, C.H., Zhao, W., Revankar, S., Speas, S., Bhagat, S., Datta, R., Phoo, C.P., Mall, U., Vondrick, C., Bala, K., Hariharan, B., 2025. Towards LLM Agents for Earth Observation. https://doi.org/10.48550/arXiv.2504.12110.

Khan, R.M., Salehi, B., Mahdianpari, M., Mohammadimanesh, F., 2021. Water quality monitoring over finger Lakes Region using Sentinel-2 imagery on Google Earth engine cloud computing platform. ISPRS Annals. V-3–2021 279–283. https://doi.org/10.5194/isprs-annals-v-3-2021-279-2021.

Lee, C., Paramanayakam, V., Karatzas, A., Jian, Y., Fore, M., Liao, H., Yu, F., Li, R., Anagnostopoulos, I., Stamoulis, D., 2025. Multi-Agent Geospatial Copilots for Remote Sensing Workflows. https://doi.org/10.48550/arXiv.2501.16254.

Li, H., Somogyi, B., Chen, X., Luo, Z., Blix, K., Wu, S., Duan, Z., Tóth, V.R., 2025. Leveraging Landsat and Google Earth Engine for long-term chlorophyll-a monitoring: a case study of Lake Balaton's water quality. Ecol. Inform. 90, 103245. https://doi.org/10.1016/j.ecoinf.2025.103245.

Liu, Y., 2025. A new human-computer interaction paradigm: Agent interaction model based on large models and its prospects. Virtual Reality & Intelligent Hardware 7 (3), 237–266. https://doi.org/10.1016/j.vrih.2025.04.001.

Liu, Z., Zhao, D., Yuan, B., 2024. RescueADI: adaptive disaster interpretation in remote sensing images with autonomous agents. https://doi.org/10.48550/arXiv.2410.13384.

NetEase Youdao. bce-embedding-base_v1: Bilingual and Crosslingual Embedding Model for RAG. Hugging Face model repository. https://huggingface.co/netease-youdao/bce-embedding-base_v1.

Pan, X., She, X., Li, X., Zhao, J., 2025. Chat-RSC: interactive remote sensing image classification via large language models. Int. J. Digit. Earth 18, 2519999. https://doi.org/10.1080/17538947.2025.2519999.

Shi, X., Gu, L., Jiang, T., Zheng, X., Dong, W., Tao, Z., 2022. Retrieval of Chlorophyll-a concentrations using Sentinel-2 MSI imagery in Lake chagan based on assessments with machine learning models. Remote Sens. 14, 4924. https://doi.org/10.3390/rs14194924.

Shu, Y., Ren, B., Xiong, Z., Paudel, D.P., Van Gool, L., Demir, B., Sebe, N., Rota, P., 2025. EarthMind: towards multi-granular and multi-sensor Earth observation with large multimodal models. https://doi.org/10.48550/arXiv.2506.01667.

Singh, S., Fore, M., Stamoulis, D., 2024. GeoLLM-Engine: a Realistic Environment for Building Geospatial Copilots. https://doi.org/10.48550/arXiv.2404.15500.

Staehr, S.U., Holbach, A.M., Markager, S., Staehr, P.A.U., 2023. Exploratory study of the Sentinel-3 level 2 product for monitoring chlorophyll-a and assessing ecological status in Danish seas. Sci. Total Environ. 897, 165310. https://doi.org/10.1016/j.scitotenv.2023.165310.

Sun, Y., Wang, D., Li, L., Ning, R., Yu, S., Gao, N., 2024. Application of remote sensing technology in water quality monitoring: from traditional approaches to artificial intelligence. Water Res. 267, 122546. https://doi.org/10.1016/j.watres.2024.122546.

Wagner, O., Gordon, J., Mousa, A., Terry, B., Baptist, J., Yetkin, O., Borges, D., Gowda, S., 2024. A generative artificial intelligence framework for Earth observation analysis. IGARSS 6986–6990. https://doi.org/10.1109/IGARSS53475.2024.10641192, 2024.

Wang, M., Zhou, C., Shi, J., Lin, F., Li, Y., Hu, Y., Zhang, X., 2025. Inversion of water quality parameters from UAV hyperspectral data based on intelligent Algorithm optimized backpropagation neural networks of a small rural River. Remote Sens. 17, 119. https://doi.org/10.3390/rs17010119.

Wang, Z., Xu, D., Khan, R.M.S., Lin, Y., Fan, Z., Zhu, X., 2024. LLMGeo: benchmarking large language models on image geolocation in-the-wild. arXiv:2405.20363. https://doi.org/10.48550/arXiv.2405.20363.

Wei, C., Zhang, Y., Zhao, X., Zeng, Z., Wang, Z., Lin, J., et al., 2025. GeoTool-GPT: a trainable method for facilitating large Language Models to master GIS tools. Int. J. Geogr. Inf. Sci. 39 (4), 707–731. https://doi.org/10.1080/13658816.2024.2438937.

Wen, C., Lin, Y., Qu, X., Li, N., Liao, Y., Lin, H., Li, X., 2025. RS-RAG: bridging remote sensing imagery and comprehensive knowledge with a multi-modal dataset and retrieval-augmented generation model. https://doi.org/10.48550/arXiv.2504.04988.

Wu, H., Shen, Z., Hou, S., Liang, J., Jiao, H., Qing, Y., Zhang, X., Li, X., Gui, Z., Guan, X., Xiang, L., 2025. AutoGEEval: a multimodal and automated evaluation framework for geospatial code generation on GEE with large Language models. ISPRS Int. J. GeoInf. 14, 256. https://doi.org/10.3390/ijgi14070256.

Yu, D., Bao, R., Ning, R., Peng, J., Mai, G., Zhao, L., 2025. Spatial-RAG: spatial retrieval augmented generation for real-world geospatial reasoning questions. https://doi.org/10.48550/arXiv.2502.18470.

Yang, F., Feng, Q., Zhou, Y., Li, W., Zhang, X., He, B., 2024. Monitoring total phosphorus concentration in the middle reaches of the Yangtze River using Sentinel-2 satellites. Remote Sens. 16, 1491. https://doi.org/10.3390/rs16091491.

Yuan, L., Mo, F., Huang, K., Wang, W., Zhai, W., Zhu, X., Li, Y., Xu, J., Nie, J.-Y., 2025. OmniGeo: towards a multimodal large Language models for geospatial artificial intelligence. https://doi.org/10.48550/arXiv.2503.16326.

Yuan, Z., Xiong, Z., Mou, L., Zhu, X.X., 2024. ChatEarthNet: a global-scale image-text dataset empowering vision-language Geo-Foundation models. https://doi.org/10.48550/arXiv.2402.11325.

Zhan, Y., Xiong, Z., Yuan, Y., 2025. SkyEyeGPT: unifying remote sensing vision-language tasks via instruction tuning with large language model. ISPRS-J. Photogramm. Remote Sens. 221, 64–77. https://doi.org/10.1016/j.isprsjprs.2025.01.020.

Zhang, T., Wei, S., Chen, S., Yu, W., Luo, M., Ji, S., 2025a. VectorLLM: Human-like extraction of structured building contours vis multimodal LLMs. https://doi.org/10.48550/arXiv.2507.04664.

Zhang, W., Cai, M., Ning, Y., Zhang, T., Zhuang, Y., Chen, H., Li, J., Mao, X., 2025b. EarthGPT-X: enabling MLLMs to flexibly and comprehensively understand multi-source remote sensing imagery. https://doi.org/10.48550/arXiv.2504.12795.

Zhang, W., Cai, M., Zhang, T., Li, J., Zhuang, Y., Mao, X., 2024. EarthMarker: a visual prompting multi-modal large Language model for remote sensing. https://doi.org/10.48550/arXiv.2407.13596.

Zheng, Y., Wang, J., Kondratenko, Y., Wu, J., 2024. Research progress in surface water quality monitoring based on remote sensing technology. Int. J. Rem. Sens. 45, 2337–2373. https://doi.org/10.1080/01431161.2024.2327086.